Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force and motion

Meaning in my words How does a net force change motion?	**How can you tell?** A cart feels a net force right. What else tells you whether it is speeding up right now?
Example or sketch Sketch a cart moving left while a net force points right. Which way does acceleration point?	**Non-example and why** “The force points right, so the cart is already going right.” What evidence is missing?

Constant net force

Meaning in my words How would you explain constant net force to a friend?	**How can you tell?** If a cart’s mass and its net force both stay fixed, what stays fixed about the motion?
Example or sketch A 2 kg cart feels 4 N right throughout two seconds. What acceleration would you label?	**Non-example and why** Does constant net force always mean steady velocity?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Zero net force

Meaning in my words How would you explain zero net force to a friend?	**How can you tell?** What happens to a moving cart's velocity if its forces add to zero?
Example or sketch Draw a cart moving right at 2 m/s a second after its horizontal forces balance and stay balanced.	**Non-example and why** Someone leaves every force off a moving cart because its net force is zero. What could still act?

Range

Meaning in my words How would you explain range to a friend?	**How can you tell?** What positions are included in 12 ± 2 m?
Example or sketch How would you mark the forecast on a number line?	**Non-example and why** A partner says only 12 m counts as a match. What do the endpoints show?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

force | acceleration | velocity | position

My marks mean

Can we release a forecast?

20

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Identify which measurements a position forecast needs.

CHOOSE ONE JOB TOGETHER

MISSION SYSTEMS ENGINEER ☐

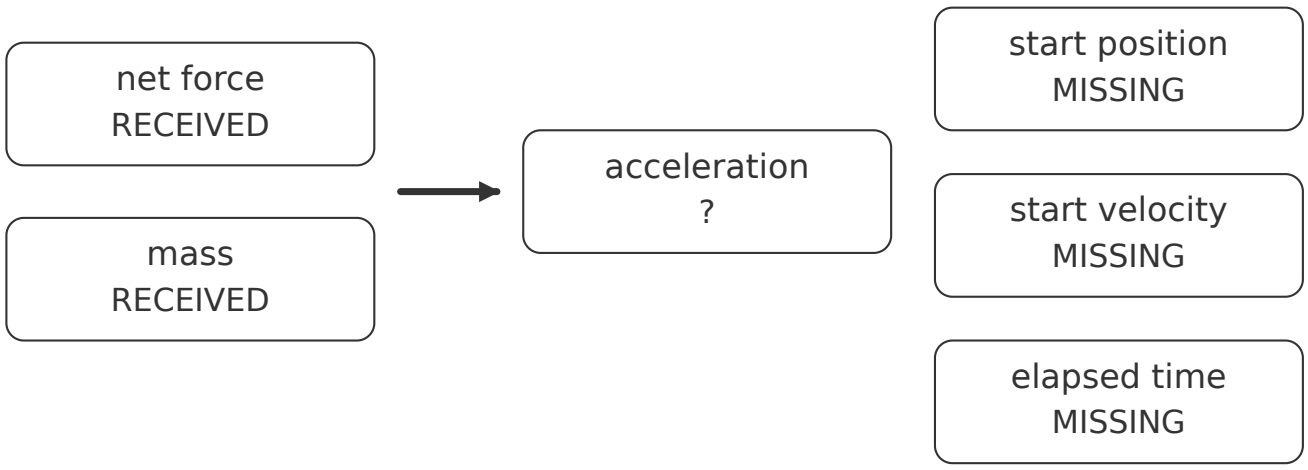
Mark what you can calculate now. Send the telescope operator a request for the measurements still needed.

NEWS ASSIGNMENT EDITOR ☐

Prepare a 'known / still needed' graphic. Tell the anchor which result can air and which must wait.

Point to a clue. Keep your first marks; add the update using the color key.

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.



01 OUR FIRST CALL

What can the desk release now? What measurements must the crew still collect?

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Is the forecast ready after the update? Say what you can release and what is still missing.

Switch speakers. What can we calculate, and what is still missing?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks